"Exploring Insights from Pizza Sales Data"

"PIZZA SALES ANALYSIS USING SQL"

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PROJECT OVERVIEW

Extract business insights using SQL queries

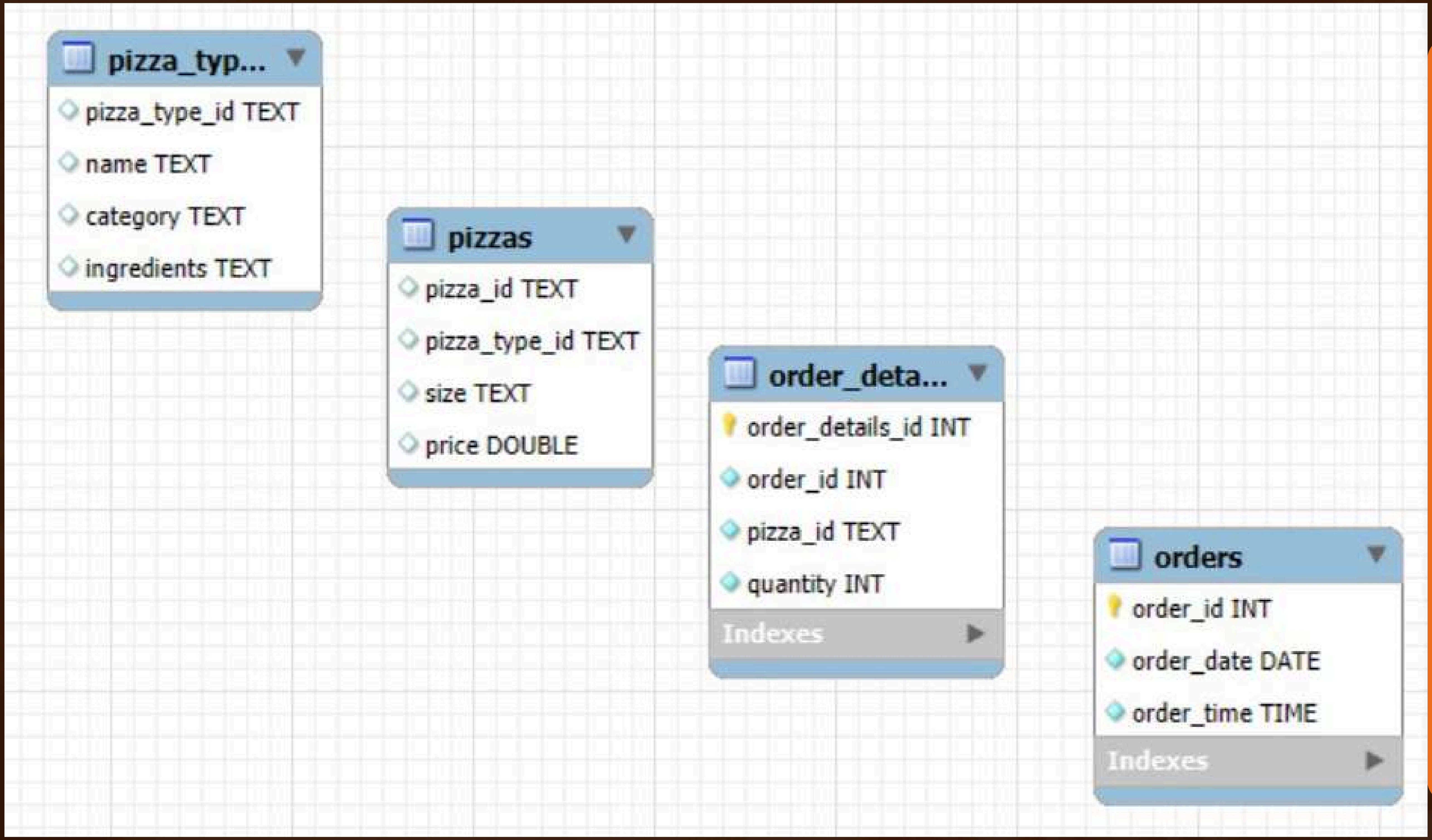
Dataset Used: Orders, Pizzas, Pizza Types, Order Details

Key Questions Answered

DATASETS

Dataset Used: Orders, Pizzas, Pizza Types, Order Details

Key Questions Answered



BASIC QUERIES & RESULTS

Retrieve the total number of orders placed.

```sql
-- Retrieve the total number of orders placed.

SELECT
    COUNT(order_id) AS total_orders
FROM
    orders
```

Result Grid

	total_orders
▶	21350

CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```sql
-- Calculate the total revenue generated from pizza sales.

SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid

total_sales
817860.05

Identify the highest-priced pizza.

```sql
-- Identify the highest-priced pizza

SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid | Filter Ro

name	price
The Greek Pizza	35.95

Identify the most common pizza size ordered.

```sql
-- Identify the most common pizza size ordered.

SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Result Grid | Filte

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
-- List the top 5 most ordered pizza types along with their quantities.--

SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
```

Result Grid | Filter Rows:

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

INTERMEDIATE QUERIES & RESULTS

Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
-- Join the necessary tables to find the total quantity of each pizza category ordered.

SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid | Filter

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Determine the distribution of orders by hour of the day.

```sql
-- Determine the distribution of orders by hour of the day.

SELECT
    HOUR(order_time), COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time)
ORDER BY HOUR(order_time);
```

Result Grid | Filter Rows:

hour(order_time)	order_count
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920

Join relevant tables to find the category-wise distribution of pizzas.

```sql
-- Join relevant tables to find the category-wise distribution of pizzas.

SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid | Filter Ro

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Determine the top 3 most ordered pizza types based on revenue.

```sql
-- Determine the top 3 most ordered pizza types based on revenue.

SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filt

avg_pizza_per_day
138

Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
-- Group the orders by date and calculate the average number of pizzas

SELECT
    ROUND(AVG(quantity), 0) AS avg_pizza_per_day
FROM
    (SELECT
        orders.order_date AS ordered_date,
            SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY ordered_date) AS order_quantity;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

ADVANCE QUERIES & RESULTS

Calculate the percentage contribution of each pizza type to total revenue.

```sql
-- Calculate the percentage contribution of each pizza type to total revenue


SELECT
    pizza_types.category,
    round(SUM(order_details.quantity * pizzas.price) / (SELECT
            ROUND(SUM(order_details.quantity * pizzas.price),
                        2) AS total_sales
        FROM
            order_details
                JOIN
            pizzas ON pizzas.pizza_id = order_details.pizza_id)*100,2 )  AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

Analyze the cumulative revenue generated over time

```sql
-- Analyze the cumulative revenue generated over time.

SELECT order_date,
    sum(revenue) OVER(ORDER BY order_date) AS cum_revenue
FROM
     (SELECT orders.order_date,
    sum(order_details.quantity*pizzAS.price) AS revenue
    FROM
    order_details
    JOIN pizzAS
    ON order_details.pizza_id =pizzAS.pizza_id
    JOIN orders
    ON orders.order_id =order_details.order_id
    GROUP BY orders.order_date) AS sales ;
```

Result Grid | Filter Rows:

order_date	cum_revenue
2015-01-01	2713.850000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21526.4
2015-01-10	23990.350000000002

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```sql
-- Determine the top 3 most ordered pizza types based on revenue for each pizza
SELECT name,revenue
FROM
(SELECT category, name ,revenue,
RANK() OVER(PARTITION BY category ORDER BY revenue DESC) AS rn
FROM
(SELECT
    pizza_types.category,
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id=pizzas.pizza_id
    GROUP BY pizza_types.category,pizza_types.name) AS a)as b
    WHERE rn <=3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75
The Sicilian Pizza	30940.5
The Four Cheese Pizza	32265.70000000065
The Mexicana Pizza	26780.75
The Five Cheese Pizza	26066.5

KEY TAKEAWAYS

Data-Driven Insights:

SQL helped uncover trends in customer ordering behavior, peak sales hours, and revenue distribution.

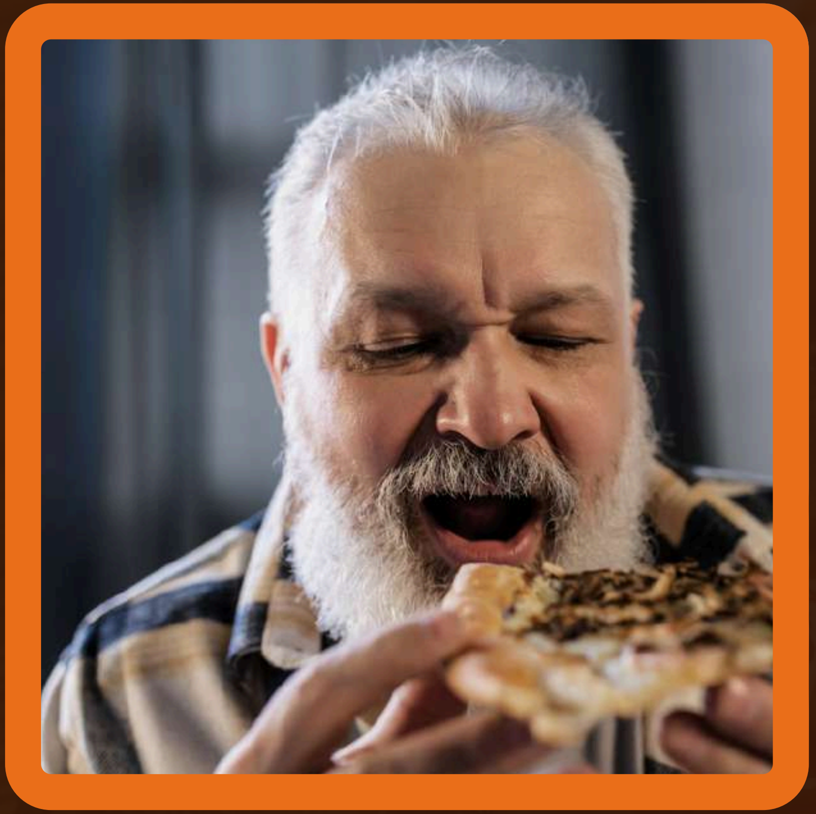
Business Value:

Understanding top-selling pizzas, revenue contribution, and demand patterns can assist in better inventory management and marketing strategies

SQL Skills Applied:

Used Joins, Aggregations, Window Functions, and CTEs to perform in-depth data analysis.

LEARNINGS & FUTURE SCOPE



Optimization:

Query performance can be improved using indexing and query tuning.

Visualization:

Future work could involve Power BI/Tableau dashboards for interactive insights.

Predictive Analytics:

Using Machine Learning to forecast future pizza sales based on historical trends.

What's Next?

Exploring more real-world datasets to strengthen SQL and data analytics skills!

Let's connect!

Feel free to share your thoughts and feedback. #SQL #DataAnalytics #BusinessIntelligence

"Pizza Sales Analysis Using SQL"

THANK YOU FOR ATTENTION